

Anoushika Vennamaneni

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AI Full Stack Engineer with 3+ years of software development experience spanning Python full-stack engineering and Generative AI applications. Experienced in building LLM-powered applications, RAG pipelines, and agentic AI workflows using Python, FastAPI, React.js, LangChain, LangGraph, and Microsoft Azure. Strong background in developing scalable APIs, cloud-native applications, and production-ready AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL, Java, C++

AI & Generative AI: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, LLM Evaluation
Backend Development: FastAPI, Django, Django REST Framework (DRF), RESTful APIs, Microservices, SQLAlchemy, Django ORM, Asynchronous APIs

Frontend Development: React.js, JavaScript, HTML5, CSS3, Responsive Web Development

Databases & Vector Stores: PostgreSQL, FAISS, Chroma

Cloud & DevOps: Microsoft Azure, Azure OpenAI Service, Azure AI Search, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD

Testing & API Tools: PyTest, Unit Testing, Integration Testing, API Testing, Postman

Security & Development Practices: JWT, OAuth 2.0, Role-Based Access Control (RBAC), API Authentication, Git, GitHub, Agile, Scrum, SDLC, Code Reviews

EXPERIENCE

AI Full Stack Engineer

IQVIA, Charlotte, Nc | Remote

Feb 2026 – Present

- Engineer production-grade Generative AI applications using Python, FastAPI, React.js, and LLM APIs for healthcare and life-sciences workflows, reducing manual information-retrieval time by approximately 35% across targeted use cases.
- Design end-to-end Retrieval-Augmented Generation (RAG) pipelines supporting 10,000+ enterprise documents, implementing document ingestion, chunking, embeddings, vector indexing, semantic retrieval, reranking, and grounded response generation.
- Develop agentic AI workflows using LangGraph and LangChain, orchestrating LLM reasoning, RAG, tool calling, enterprise API interactions, conversation memory, and human-in-the-loop approval for multi-step business processes.
- Build asynchronous Python/FastAPI microservices and REST APIs for AI inference, document processing, retrieval, streaming LLM responses, and enterprise integrations, supporting 5,000+ API requests per day across application workflows.
- Implement prompt engineering, context management, response validation, grounding, fallback strategies, and LLM evaluation, improving acceptable response rates from approximately 78% to 91% across internally evaluated question sets.
- Develop reusable React.js AI interfaces and dashboards featuring streaming responses, document-grounded Q&A, source references, conversation history, user feedback, and workflow visualization.
- Containerize and deploy AI services using Docker, Kubernetes, and Microsoft Azure, integrating CI/CD, application monitoring, logging, API security, and PostgreSQL persistence to support reliable production releases.

Environment: Python, FastAPI, React.js, JavaScript, REST APIs, Microservices, Generative AI, LLMs, LangChain, LangGraph, RAG, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, PostgreSQL, Docker, Kubernetes, Microsoft Azure, CI/CD, Git, GitHub

Graduate Assistant

Youngstown State University, Youngstown, OH

August 2024 – August 2025

- Supported university-level software and research projects using Python and React.js, extending prior full-stack development experience while gaining hands-on exposure to Generative AI and LLM application development.
- Developed 3 AI-enabled academic prototypes, including a document question-answering application using Python, FastAPI, React.js, and LLM APIs to explore practical AI-assisted information retrieval.
- Implemented prototype RAG workflows using document extraction, chunking, embeddings, FAISS/Chroma, and semantic retrieval to generate context-aware responses from academic and reference documents.
- Experimented with 30+ prompt variations and test scenarios using LangChain and prompt engineering techniques to evaluate context handling, response relevance, and conversational behavior.

- Built 10+ Python/FastAPI REST endpoints and reusable React components supporting document upload, processing, retrieval, conversational interactions, and application data management.
- Used PostgreSQL, Docker, Microsoft Azure, Git, and GitHub for application data management, reproducible development environments, source control, deployment, and documentation of project outcomes.

Environment: Python, FastAPI, React.js, JavaScript, REST APIs, Generative AI, LLM APIs, LangChain, RAG, Prompt Engineering, Embeddings, FAISS, Chroma, PostgreSQL, Docker, Microsoft Azure, Git, GitHub

Full Stack Developer

Wunderman Thompson Commerce, Noida, India

May 2022 – November 2023

- Developed and maintained Python/Django full-stack commerce applications for 4 client brands, supporting customer-facing storefronts serving more than 50,000 monthly active users.
- Designed and integrated 20+ RESTful APIs using Python, Django, and Django REST Framework for product catalogs, shopping carts, checkout, customer accounts, order processing, and third-party services.
- Built 25+ reusable React.js components for product discovery, search, checkout, and account-management workflows, improving frontend consistency and reducing duplicate development effort.
- Designed PostgreSQL data models and optimized Django ORM queries, improving response times for selected high-traffic application operations by approximately 25%.
- Implemented JWT/OAuth 2.0 authentication, RBAC, input validation, and PyTest-based unit/integration testing, increasing backend test coverage from approximately 45% to 70%.
- Containerized applications using Docker and Kubernetes and supported automated deployments through Microsoft Azure and Jenkins CI/CD, reducing manual release steps while collaborating within a 6-member Agile engineering team.

Environment: Python, Django, Django REST Framework (DRF), React.js, JavaScript, HTML5, CSS3, REST APIs, PostgreSQL, SQL, Django ORM, JWT, OAuth 2.0, PyTest, Docker, Kubernetes, Microsoft Azure, Jenkins, CI/CD, Git, GitHub, Agile/Scrum

PROJECTS

AI Customer Support & Ticket Resolution Platform

- Developed an agentic AI customer-support platform using Python, FastAPI, React.js, LangGraph, and LangChain to analyze tickets, retrieve knowledge, generate responses, and assist with request routing.
- Orchestrated multi-step AI agent workflows combining LLM reasoning, RAG, semantic retrieval, tool execution, and human-in-the-loop approval.
- Built REST APIs and a React dashboard for ticket management, AI recommendations, agent review, and workflow tracking, with PostgreSQL persistence and containerized deployment on Microsoft Azure.

Tech Stack: Python, FastAPI, React.js, LangGraph, LangChain, LLM APIs, RAG, FAISS/Chroma, PostgreSQL, Docker, Microsoft Azure

AI Job Application Analytics Dashboard

- Developed a full-stack AI job application platform using Python, FastAPI, React.js, and PostgreSQL to track applications, interview stages, job descriptions, and recruiting activity.
- Integrated LLM-powered resume-to-job matching and job-description analysis to extract required skills, identify skill gaps, and generate personalized match insights.
- Built analytics APIs and React visualizations for application status, skill alignment, interview progress, and job-search trends, with containerized deployment on Microsoft Azure.

Tech Stack: Python, FastAPI, React.js, PostgreSQL, LLM APIs, Prompt Engineering, REST APIs, Docker, Microsoft Azure, Git

EDUCATION

Youngstown State University, Youngstown, OH

Jan 2024 – Dec 2025

Master of Science in Computer Science

CERTIFICATIONS

- [AWS Certified Developer - Associate](#)